

Phyling

Technical Data Sheet: 3D Instrumented Pedals (PI3D)
Three-dimensional analysis of pedalling forces and dynamics.

1. Force Measurement (3 axes)

Analysis of propulsive and non-propulsive (medio-lateral) forces.

Characteristic	Technical Specifications
Sensor type	3-axis strain gauges (Fx, Fy, Fz)
Measurement range (Fx, Fy)	$\pm 100\text{ kg}$ ($\approx 1000\text{ N}$)
Measurement range (Fz)	$\pm 450\text{ kg}$ ($\approx 4500\text{ N}$)
Accuracy (mean error)	$0.5 \pm 1.3\%$
Resolution	0.001 % of measurement range
Sampling rate	200 Hz
Calibration	Factory (transverse sensitivity matrix)

2. Kinematics & Reference Frame

Tracking of pedal orientation and cadence for power calculation.

Characteristic	Specifications
Orientation	Built-in gyroscope
Cycle detection	Magnetic “Top Tour” (once-per-revolution) sensor
Cadence	Instantaneous measurement (RPM)
Calculated data	Power (W), pedalling efficiency, L/R balance

3. Physical Characteristics

Designed for bike fitting and intensive use on a test bench or on the road.

Characteristic	Details
Materials	High-strength aluminium
Weight	530 g (pair)
Dimensions	97 × 65 × 59 mm
Cleat compatibility	Look Keo, Shimano SPD-SL, Platform
Q-Factor	53 mm
Stack height without cleat	Look 16.9 mm Plate 16 mm Shimano 14.6 mm
Ingress protection (IP) rating	No
Operating temperature	0°C to +40°C

4. Connectivity & Power

Recording flexibility for research or real-time fitting.

Characteristic	Value
Communication	Bluetooth Low Energy (BLE)
Local storage	SD card included (standalone recording)
Data retrieval	USB (cable included)
Battery life	2 hours (continuous streaming)
Charging	Magnetic connector
Charging time	2 h

5. Maintenance & Software

Characteristic	Details
Interface	Phyling app (real-time display via Maxi-Phyling)
Analysis	Raw data export to CSV/Excel via the Phyling app
Maintenance	Annual calibration recommended
Warranty	1 year parts and labour